

Optimal Sensor Placement in Smart Sewer Systems Using Network Topology and Elevation

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Abstract: Smart sewer systems consisting of water level sensors can effectively detect overflow events that may present a threat to water quality, human health, and the environment. Designing the optimal layout of the sensor with the minimum number of sensors is essential to minimize the cost and maximize the coverage. This paper proposes a novel methodology to achieve the aim by combining genetic algorithms (GAs) with a blockage detection condition for sensor placements. The blockage detection condition that relies on network topology and elevation information is used to identify the pipe coverage capacity of possible sensor locations. This information serves as input data for the optimization process, which utilizes a GA to identify the optimal sensor placement. The effectiveness of the methodology is verified through its applications to two real gravity sanitary sewer systems with distinct network topologies and elevation information. The results demonstrate that the proposed method can effectively determine the optimal number and placement of sensors required to cover the entire sewer systems throughout multiple investment phases. This contributes to enhancing the effectiveness of investment strategies for smart sewer systems. DOI: 10.1061/JWRMD5.WRENG-6660. © 2025 American Society of Civil Engineers.

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Introduction

A sewer system consists of a network of pipes and associated infrastructures designed to transport wastewater from collection points to treatment plants efficiently and safely. Despite its essential function, the system can pose potential hazards to both humans and the environment, particularly due to overflow events that are caused by unforeseen blockages in dry weather (NRMCC 2004). Blockages occur when pipes are either completely or partially blocked due to external factors (e.g., construction or land subsidence) or internal factors (e.g., soil or tree roots entering the pipes through displaced pipe joints or cracks in the pipes) (Walski and Barnard 2004). Over time, the blockage can get worse as solids get caught in the obstruction, leading to sewer overflow through manholes. This poses a challenge for sewer system managers to detect blockage development in sewer pipes and give an early warning for overflow prevention.

One of the conventional methods to manage blockages and sewer overflow in sewer systems is a combination of reactive and preventative approaches. While the reactive approach is often implemented in response to customer complaints, the preventative approach involves planned activities, such as regular cleaning and inspections. In terms of sewer inspection techniques, using closed-circuit television (CCTV) cameras and using acoustic devices are the most common ways of inspecting the interior conditions of a sewer pipe including identifying any blockages as it is cost-effective and nondestructive (Do et al. 2023). However, the limitation of these techniques is that they are unable to continually monitor and proactively detect potential blockages in large and complex networks (Duran et al. 2002; Romanova et al. 2013).

With the increasing development of the Internet of things technology in recent years, the use of sensors for the early detection of damage in water networks has been increasingly feasible and practical. Unlike CCTV and traditional sewer acoustic techniques, the use of sensors focuses on detecting early abnormalities based on the time series data recorded, allowing for proactive alterations that can prevent potential system failures. The effectiveness of using sensors in water distribution systems has been demonstrated in many applications. For example, pressure and flow sensors have been used for pipe leak and crack detections (Stephens et al. 2020; Zeng et al. 2020; Zhang et al. 2020), and water quality sensors have been applied to detect contaminant events in both sewer and water distribution systems (Giudicianni et al. 2020; Nourinejad et al. 2021; Shastri and Diwekar 2006; Yazdi 2018).

Water utilities have embarked on the development of smart sewer systems that incorporate sensor networks to effectively manage and mitigate overflow events. One such system is the smart sewer monitoring system implemented in the urban areas in South Australia, where distributed sensors (e.g., ultrasonic water level sensors) have been installed in sanitary sewer systems to monitor blockages (Do et al. 2023). The smart sewer systems can effectively detect overflow events, but they have a shortcoming of the high

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capital cost for the sensing system. Therefore, installing sensors at every node in a sewer network is not a realistic option (Adedoja et al. 2018; Yazdi 2018). Moreover, a smart monitoring system is often implemented in multiple phases due to budget limitations, leading to the need for priority locations of a fixed number of sensors for each investment phase. As a result, it is essential to minimize the cost while maximizing the capacity of each sensor by identifying an optimal number and locations of sensors. This issue has attracted the attention of both academia and industry (Adedoja et al. 2018).

While optimizing sensor placement for proactive detection of sewer overflow in sewer systems has not yet been found in the literature, it has been well studied in different areas across the water-related field, such as water distribution systems (Hart et al. 2010; Naserizade et al. 2018; Shastri and Diwekar 2006; Zecchin et al. 2022), hydrology (Fattoruso et al. 2015; Ogie et al. 2017), or river systems (Welch et al. 2013; Zubeizu et al. 2020). Many techniques have been developed to optimize the sensor locations, including both simple and complex optimization models and algorithms. For instance, a new stochastic programming approach was proposed by Shastri and Diwekar (2006) to solve uncertainties in the water distribution network during the process of determining the optimal sensor location. Hart et al. (2010) developed a nonlinear sensor placement formulation, which is optimized using a linear approximation and then solved by a mixed-integer linear program. Complex algorithms such as a heuristic method (Rathi and Gupta 2016) or greedy heuristics and convex relaxation (Zhao et al. 2016) were developed and used to optimize water sensor placements. In another paper, a nonlinear integer program and heuristic algorithm were combined to determine the optimal number of sensors and their location to detect SARS-Cov-2 in the sewer system (Nourinejad et al. 2021). Five different procedures consisting of both single and multiple objectives for the optimal sensor placement problem were also compared and evaluated by Banik et al. (2017).

Genetic algorithms (GAs) are among the most popular optimization techniques. GAs have been successfully applied in various engineering optimization problems including the design and operation of water distribution networks (Do et al. 2016; Goldberg and Kuo 1987) and sewer system management (Afshar 2012; Al-Obaidi et al. 2017; Parker et al. 2000). The robustness of genetic algorithms and their effectiveness at efficiently searching large solution spaces have been demonstrated both theoretically and empirically (Simpson et al. 1994). In the last decade, GAs have been used for sewer overflow reduction optimization in combined sewer systems (Joseph-Duran et al. 2014; Ogidan and Giacomoni 2016; Rathnayake and Tanyimboh 2015; Zimmer et al. 2015).

Optimization techniques are typically coupled with hydraulic models to simplify the system representation (Hart et al. 2010; Shastri and Diwekar 2006). This combined approach is instrumental in addressing the complex internetwork effects and making more informed decisions about the design and operation of the water distribution system. The integrated approach can demonstrate good performance if a full hydraulic model is available. However, obtaining a fully calibrated hydraulic model in sewer systems can be challenging because of data availability (Adedoja et al. 2018; Giudicianni et al. 2020; Nourinejad et al. 2021). To reduce reliance on hydraulic models, researchers have begun focusing on the use of system characteristics (e.g., network topology) to enhance sensor placement optimization strategies (Giudicianni et al. 2020; Saldarriaga et al. 2021; Simone et al. 2022) and address other issues, including vulnerability assessment and monitoring system planning (Simone et al. 2023).

The topology and elevation are the main factors impacting the structure and operation of sewer systems, especially gravity sewer systems. They decide how the different components are connected and how sewer flows through the system based on gravity. With such an important role, exploring the characteristics of the network topology and elevation may support water utilities in finding the most suitable sensor placement in the system without using hydraulic models. In the literature, topological properties of the network have been used by Giudicianni et al. (2020) to identify suitable placements of quality sensors in the water distribution networks. Instead of using hydraulic information, the study utilized the concept of community structure in complex water distribution networks, grouping similar topological features into clusters. Sensor placement was then focused within each cluster to ensure complete network coverage with sensor positions determined using topological centrality metrics. Similarly, the network topology and land topography were also used as criteria to select an initial layout before optimizing sewer network design using dynamic programming (Saldarriaga et al. 2021). However, the network topology and catchment slope have never been exploited to optimize sensor coverage in the gravity sewer system to detect blockages.

This paper proposes a novel methodology based on the combination of genetic algorithms and a blockage detection condition for the sensor locations using network topology and elevation information to address two main objectives: (1) estimating the number of sensors required for full gravity sanitary sewer system coverage; and (2) determining optimal locations of a fixed number of sensors. This methodology is then applied to two real sanitary sewer systems. It is expected to improve predictive management in sewer systems by reducing cost and computational effort without relying on hydraulic models.

This paper is structured as follows: first, the characteristics of the gravity sewer network and the performance of water level sensors are discussed to identify the blockage detection conditions of the sensors in the network. A blockage detection condition for potential sensor locations is then introduced and applied to find both covered nodes and uncovered nodes associated with each potential sensor location. Next, optimization problems are formulated to minimize the number of sensors for the entire system and determine the optimal locations of sensors subject to the given number of sensors for an investment phase. Case studies are introduced before showing the results and discussions of the application of the proposed methodology. Finally, the conclusions and suggestions for future work are given.

Sewer Monitoring Systems with Water Level Sensors

A sewer system includes various elements such as sewer pipes (main, branch, and lateral), manholes or access structures, odor scrubbing systems, treatment facilities, and vent shafts. However, to optimize sensor placement, the focus will primarily be on the sewer pipes and manholes. The main components of sewer systems can be effectively represented using graph theory, involving nodes and links (N, L). The nodes (including manholes, junctions, and outlets) and links (pipes) are represented by the vertices ($N = 1, 2, \dots, n$) and edges ($L = \lambda_1, \lambda_2, \dots, \lambda_n$), where $\lambda_j = (i, k)$ represents the link between node $i \in N$ and node $k \in N$. An example of a sewer system graph is shown in Fig. 1(a).

Manholes are access points in the sewer system, including maintenance holes and inspection holes. While maintenance holes allow personnel and equipment access to the sewer system, inspection holes only provide access to equipment. Therefore, to ensure strict

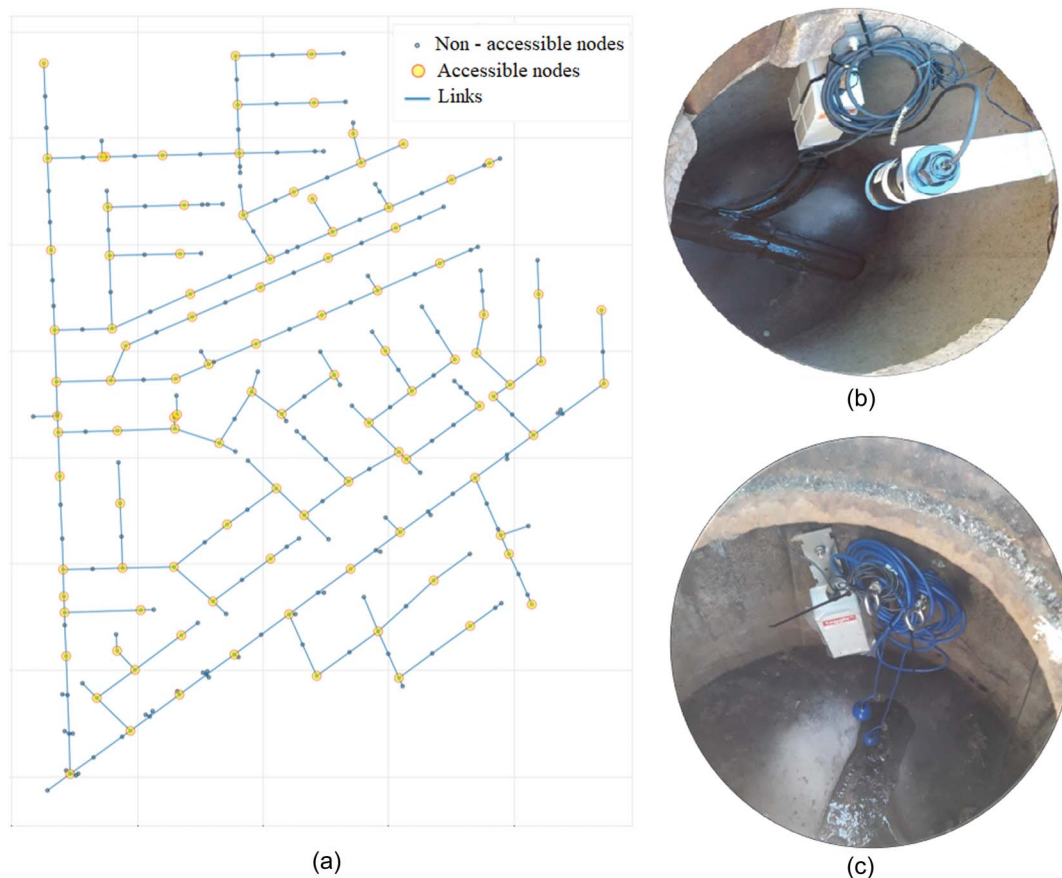


Fig. 1. Sewer system and level sensors: (a) graph representation of sewer system; (b) ultrasonic sensor; and (c) float sensor deployment at maintenance holes.

work health and safety conditions, maintenance holes are considered a safe place for sensor installations in sewer systems. As loops are not allowed in sewer systems, overflow events can only occur at upstream manholes of the blockage.

The ultrasonic level sensor in Fig. 1(b) is located near the top of the maintenance hole and vertically directed to the sewer stream. Water levels are measured every 5 min and transmitted to a database instantaneously. In Fig. 1(c), two float sensors are set corresponding to two levels of low alarm (150 mm) and high alarm (300 mm). Under normal conditions (i.e., water levels are below the alarm levels), the float sensors will transmit a “healthy pulse” every hour. This transmission is set up to ensure that the sensors are operational and signals can receive. When entering an alarm state, the sensors will transmit data, either low or high-level alarm, every 15 min.

Sensors can detect blockages by observing abnormal hydraulic conditions. For example, a persistent low water level may indicate a blockage occurring upstream of the sensor, while a persistent high water level indicates a blockage occurring downstream of the sensor. However, detecting a blockage informing upstream of the sensor by observing the drop in water levels is challenging in practice. This is because sensors are installed at the maintenance holes that are often the connection points of different sewer branches. In addition, the sewer discharge from households above the maintenance hole is often intermittent, which may result in a very low flow or zero flow during off-peak hours. Another reason is that data collected from the sensor are either low-frequency or nonsynchronized. Fig. 2 plots an example of water level at the upstream and downstream of a blockage event. While a clear signal

of a blockage was detected by the sensor located upstream of the blockage, there is an unclear indication of a blockage from the sensor located downstream of the blockage. Thus, the proposed method in this study focuses only on the placement of sensors to identify blockages at their downstream pipes.

Proposed Blockage Detection Condition for Water Level Sensor Locations

Based on the topological properties and elevations of gravity sewer systems, a blockage detection condition for a water level sensor location is proposed that the sensor must observe the rise in water level at the sensor location caused by a blockage in a downstream pipe and before any overflow occurs at the downstream nodes of the sensor location. This means that during a blockage event in Pipe 0, the water levels will rise above the typical level observed under normal conditions [Fig. 3(a)] at the sensor location (M3) and its downstream nodes (M0, M1, and M2) as shown in Fig. 3(b). Within this condition, the upstream nodes and pipes relative to the sensor location are not taken into consideration and are referred to as uncovered nodes and links. The condition can be expressed by the equation

$$Z_0 + H_d + \epsilon > Z_s + H_s \quad (1)$$

where Z_0 = invert elevation of the downstream nodes of a sensor location; H_d = water depth threshold at the downstream nodes that can trigger a potential overflow alert; H_d = percentage of the depth of the downstream nodes (H); and H_d will be determined by water

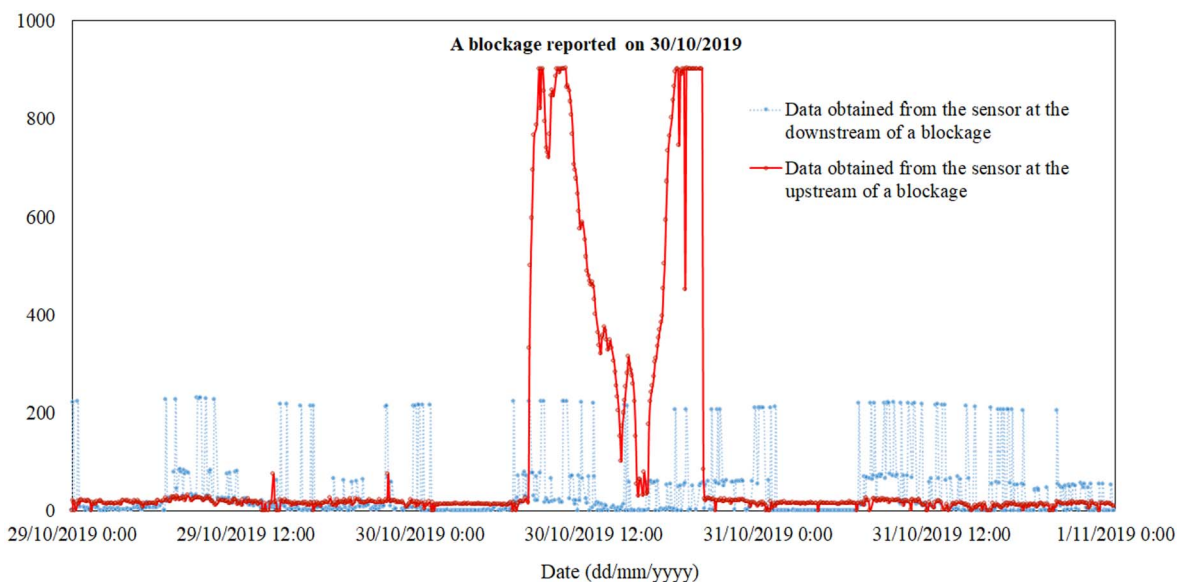


Fig. 2. Data recorded by upstream and downstream sensors of a blockage event. It shows no sign of the blockage from the sensor installed downstream of the blockage.

utilities through a trade-off analysis involving the target of preventing sewer overflows and the capital investment required for smart sewer systems. The lower the H_d , the higher the associated cost, given the increased need for sensors. In this research, H_d is set at 70% of H based on the rise of water level detected by sensors during blockage events in two relevant sewer systems. ε is the error

associated with the surface elevations. Z_s is the invert elevation of maintenance hole s ($s = 1, 2, 3$, and 4, as shown in Fig. 3) where the sensor is located. H_s is the typical water depth (i.e., no sudden discharge) at maintenance hole s .

Water levels in sewer systems typically remain below the pipe diameter [Fig. 3(a)]. To demonstrate the blockage detection condition, a blockage is assumed to occur in Pipe 0 [Fig. 3(b)], potentially leading to an overflow event at inspection hole M0. In this system, sensors can be installed at all maintenance holes M1, M2, M3, and M4. As a result of the blockage, the water level rise in the pipes can only reach maintenance holes M1, M2, and M3 through Pipes 1, 2, and 3, while the water level at M4 remains in its normal conditions. In such cases, only sensor locations at M1, M2, and M3 meet the blockage detection condition outlined in Eq. (1), but redundancy is observed for the sensors at M1 and M2. This is because the installation of the sensor at M3 is sufficient to cover nodes M0, M1, and M2, as well as Pipes 0, 1, 2, and 3. In this context, we refer to M0, M1, and M2 as downstream-dependent nodes (DD nodes) of M3 while M0 is a "downstream-independent node" (DI node) of M4. It should be noted that a DD node must satisfy Eq. (1); otherwise, it will be categorized as DI nodes.

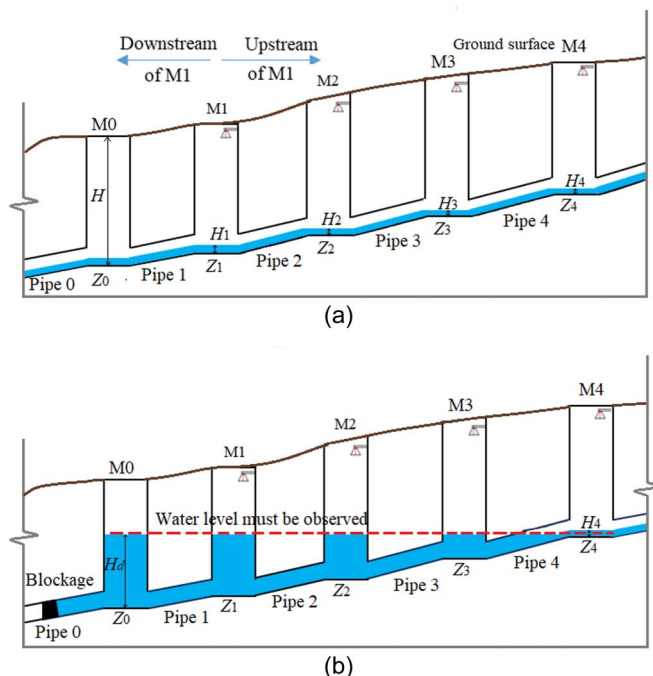


Fig. 3. Example of the blockage detection condition for sensor locations. Water level in (a) normal conditions; and (b) blockage conditions. In this example, sensors at M1, M2, and M3 can detect water level rise at an inspection hole M0 caused by a blockage occurring at Pipe 0, while the sensor at M4 cannot detect the blockage. M3 is the optimal location to install a sensor as it is the most distant sensor that can observe this blockage. Sensors at M1 and M2 are redundant in this context.

Identification of DD and DI Nodes

To identify DD and DI nodes, the required input data include the elevations of nodes and manholes, the height of the maintenance holes, the starting node, and the ending node of each link of the system, flow directions, and branching networks. A breadth-first search (BFS) algorithm is first applied to identify all possible downstream nodes of every maintenance hole. BFS is a graph traversal algorithm that explores all the nodes of a graph by visiting all the neighboring nodes of the starting node before moving on to further nodes (Moore 1959). Intersection points (such as Points 5 and 10 in Fig. 4) in gravity systems can be treated like any other node. When reaching an intersection, it continues exploring the downstream connections for all unvisited neighbors. If there are multiple downstream connections at an intersection, BFS will add all of them to the queue for further exploration.

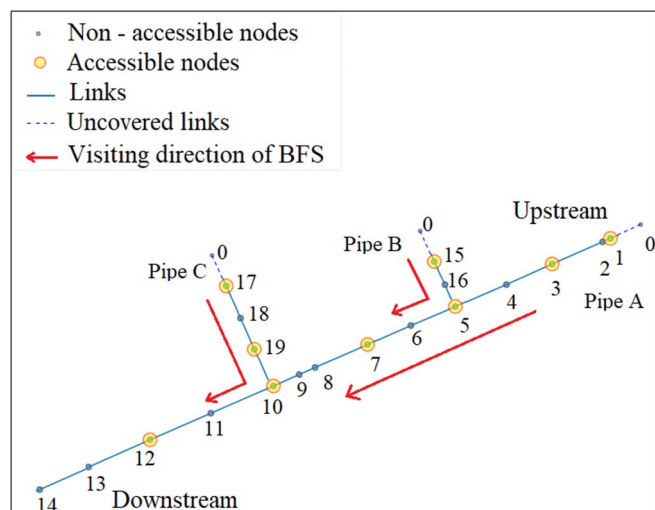


Fig. 4. Example of the breadth-first search (BFS) algorithm.

An illustration of the BFS algorithm is depicted in Fig. 4. In the diagram, Node 0 represents an upstream inspection hole that could not be covered by any sensor. As a result, the links from Node 0 to Nodes 1, 15, and 17 are uncovered (dashed line). In Pipe A, the BFS commences from Node 1 (the first potential location to place a sensor), which serves as the source node, and subsequently visits the downstream nodes in the following order: 2, 3, 4, . . . , 14. At each visited downstream node, Eq. (1) is used to determine whether it is a DD node or a DI node. To simplify the computation process, the error ε in Eq. (1) is set to 0. The BFS is applied to all pipes in the system. **After completing the calculations, all downstream-dependent nodes are recorded and used as input data (Table 1) for optimization procedures.**

Formulation of Optimization Problems

Genetic Algorithm

A genetic algorithm is a type of search algorithm that relies on artificial evolution, as described by Holland (1975). **This algorithm is designed to imitate the rules of natural selection and population genetics to achieve adaptation. In a genetic algorithm, a population of individuals (solutions) is evolved through multiple generations using processes such as selection, crossover, and mutation. The selection process is to determine which individuals from the current population will be selected to generate the next population. Crossover is used to explore the search space and find optimal solutions**

Table 1. Downstream-dependent nodes found by BFS and the blockage detection condition

Accessible nodes	Number of DD nodes	List of DD nodes
1	5	2, 3, 4, 5, 6
3	4	4, 5, 6, 7
5	6	6, 7, 8, 9, 10, 11
7	5	8, 9, 10, 11, 12
10	3	11, 12, 13
12	2	13, 14
15	3	16, 5, 6
17	4	18, 19, 10
19	2	10, 11

by swapping or combining parts of the chromosomes of the parent solutions to generate a new offspring solution. **The mutation is applied to prevent local optimum solutions.**

To apply the GA to a specific problem, the problem-dependent components of the algorithm (including the encoding and evaluation functions) should be defined. The encoding function is used to convert solutions into a form that can be manipulated by the algorithm, while the evaluation function (fitness) is used to assess the quality of solutions and guide the search process toward better solutions. These functions are specifically defined for two optimization problems in this paper as follows.

Encoding Function and Fitness Function for the Entire Network Coverage

The first optimization problem is to determine the minimal number of sensors for the entire network coverage, where the decision variables represent the locations at which sensors should be installed to cover the system. In addressing this optimization problem, the GA employs a binary encoding function, where each individual in GA (i.e., a solution) is represented by a binary string. Each bit in the string is aligned with a potential sensor location at a manhole. The length of the string corresponds to the total number of manholes available for sensor placement. Within this binary string, each '0' indicates that a particular manhole does not have a sensor installed, while each '1' confirms the presence of a sensor at that location. Consequently, the total number of installed sensors is determined by counting the number of '1's in the binary string. For instance, a binary string representing a sewer network with 10 manholes, such as [0,1,1,0,0,1,0,0,1,0], indicates that sensors are installed at Manholes 2, 3, 6, and 9. This method effectively maps the sensor installation across the network, facilitating the optimization of sensor placement for comprehensive network coverage.

The fitness of each individual is calculated relying on the number of downstream nodes observed by installed sensors as shown in Eq. (2). The GA is trying to search for solutions in which all downstream nodes in the system are observed by sensors. This is satisfied if the total set of downstream nodes is a subset of the union of all the sets of downstream nodes observed by each sensor. In this optimization problem, the best solution will have the smallest fitness

$$\text{Fitness} = \begin{cases} \sum_{i=1}^{n_s} n_{d,i} & \text{if } A \subseteq \bigcup_{i=1}^{n_s} D(i) \\ \text{Penalty} & \text{otherwise} \end{cases} \quad (2)$$

$$\text{Subject to } 0 \leq n_s \leq M$$

where n_s = number of installed sensors; $n_{d,i}$ = number of downstream nodes covered by sensor i ; $D(i)$ = set of downstream nodes observed by sensors i ; and A = set of unique downstream nodes in the system (calculated). $A = N - N_u$, where N is the set of all nodes of the system and N_u is the set of upstream nodes that could not be observed by any sensor. In Fig. 4, $N = \{0, 0, 0, 1, 2, 3, \dots, 19\}$, $N_u = \{0, 0, 0\}$, and $A = \{1, 2, 3, \dots, 19\}$. $\bigcup_{i=1}^{n_s} D(i)$ is the union of all the sets of downstream nodes observed by installed sensors, including duplicated nodes that are covered by at least two sensors, and M is the number of manholes in the system. A penalty will be applied if any downstream node is not observed. **In the case of minimization problems, a substantial penalty value that is greater than the number of nodes in the sewer system is essential to force the genetic algorithm to prioritize feasible solutions and prevent premature convergence. The penalty value is determined based on the number of nodes in the real sewer system.**

Encoding Function and Fitness Function for a Limited Number of Sensors Given in an Investment Phase

It is optimal to install sensors throughout the entire sewer system simultaneously. However, due to budget constraints, the implementation of a smart sewer system may occur in various phases. Therefore, the second optimization problem in this study is to find the optimal locations for a limited number of sensors given in an investment phase of a smart sewer system, which not only maximizes the effectiveness of each phase but also ensures full coverage for the sewer systems after completing the final phase. In other words, this problem is to find the maximum total length of downstream pipes that the given sensors can observe and detect the water level rise caused by blockages. **The decision variables of this optimization are the locations of sensors, so the encoding function, in this case, is value encoding. In this encoding scheme, each individual is depicted as a vector of integer values, with each value representing a numerical order for the manhole (i.e., sensor location).**

To find the optimal locations for a fixed number of sensors, GA computes the total length of pipes covered by selected manhole locations in each individual in the population and then evaluates the fitness of each individual. A constraint in the procedure for a limited number of sensors is that each sensor installed must fully cover at least one pipe in the sewer system. The GA will seek an optimal solution that satisfies the minimum of the fitness function. The fitness function is selected in the following form:

$$\text{Fitness} = \begin{cases} \frac{1}{\sum_{j=1}^m l_{\lambda_j}}, & \sum_{j=1}^m l_{\lambda_j} > 0, \lambda_j \in L_{dd} \\ \text{penalty}, & \text{otherwise} \end{cases} \quad (3)$$

where m = total number of pipes covered by given sensors and linked by two DD nodes; and λ_j = pipe j that is observed by a given sensor. $\lambda_j = (i, k)$, $(i, k) \in V$, where i and k are upstream and downstream nodes of pipe λ_j and V is the set of unique downstream nodes covered by given sensors. L_{dd} is the set of downstream pipes observed by installed sensors. $L_{dd} = \{\lambda_1, \lambda_2, \dots, \lambda_m\}$. l_{λ_j} is the length of pipe λ_j . A penalty is added to ensure that the solutions that fail to cover any pipes are handled appropriately. In this method, the penalty can be a fixed value determined by the characteristics of the sewer system.

Selecting Methods for Sensor Locations

The selection of optimal deployment locations becomes crucial, especially considering the limited number of sensors available for each investment phase. The optimal location selection for an investment phase can be achieved through a simultaneous selection method (i.e., selecting all locations at the same time). For example, when utilizing 10 sensors, the simultaneous selection method considers all 10 potential sensor locations simultaneously, evaluating their observation capacity based on the maximum total length of pipes covered by the sensors. The simultaneous method always provides the global optimum during the investment stage of a smart sewer system.

Additionally, a multistage selection approach is employed to achieve optimal sensor location selections for a system with multiple investment phases. This approach involves the sequential evaluation and selection of the best sensor locations for each investment phase. In a sewer system with five investment phases and 10 sensors for each phase, the multistage selection approach initially assesses and selects the best combination of 10 sensors for the first investment phase. After determining the optimal locations for the first phase, sensor locations for the subsequent phase are chosen and

evaluated, considering the information obtained from the selected locations in the previous phase. This process continues until sensors for the final investment phase are selected. The multistage selection method not only enables the identification of the best locations for each investment stage but also ensures a near-optimal global solution, facilitating quicker decision-making.

Case Studies

Two smart sanitary sewer systems that are scheduled for the extension are used to prove the efficacy of the proposed method. It is noted that the elevations of these systems are significantly different. While **Sewer System I is quite flat, the second one is steep.** The impact of elevations on optimizing the sensor placement will be evaluated in the result and discussion section. Both sewer systems are gravity-driven with a diameter of 0.150 m for pipes except for an outlet pipe of Sewer System I with a diameter of 0.225 m. The materials of pipe systems are vitreous clay and unplasticized polyvinyl chloride. In these systems, the diameter of the inspection holes is 0.15 m, while the diameter of the maintenance holes is 0.6 m. Some basic information of these systems is shown in Table 2.

Sewer System I consists of 266 pipes with a total length of about 12.5 km. Located in a flat area, this system has a quite low slope, with an average pipe slope of approximately 1.4%. It has a total of 103 maintenance holes, with an average depth of 2.7 m, of which 10 maintenance holes had been installed with sensors. In Sewer System II, there are 189 maintenance holes with a depth in a range of 0.7–4.1 m. This system has a total pipe length of 11.3 km for 322 pipes and is currently installed with 82 ultrasonic water level sensors.

Fig. 5 shows an application of the proposed method to these sewer systems. The blockage detection condition is applied to determine DD nodes and independent nodes (ID nodes) of each manhole in both sewer systems. The GA is then used to minimize the number of sensors for these systems based on the list of DD and ID nodes. In this case, the impact of elevation on the minimum number of sensors for each sewer system is studied. The objective of determining optimal locations for a fixed number of sensors is explored in Sewer System I, where only 10 sensors have been installed. The insight is deemed crucial by the water utility for informing their upcoming investment phase.

In terms of selecting GA parameters, Hassanat et al. (2019) suggested that a larger population size tends to yield better results compared to a smaller one, and crossover was more effective than the mutation in the larger population size. Thus, we chose a large population size and crossover with small values for mutation and generation to address the objectives of this paper. Through experimentation with various parameter sets, we determined the following values for the GA parameters in the case studies: a population size of 1,000, 200 generations, a crossover probability of 0.75, a mutation probability of 0.06, and penalty values set at 1,000.

Table 2. Information on two gravity sewer systems

Parameter	Sewer System I	Sewer System II
Slope of pipe (%)	1.4	7.8
Total length (m)	12,500	11,300
Number of maintenance holes	103	189
Number of pipes	266	322
Number of nodes	267	323

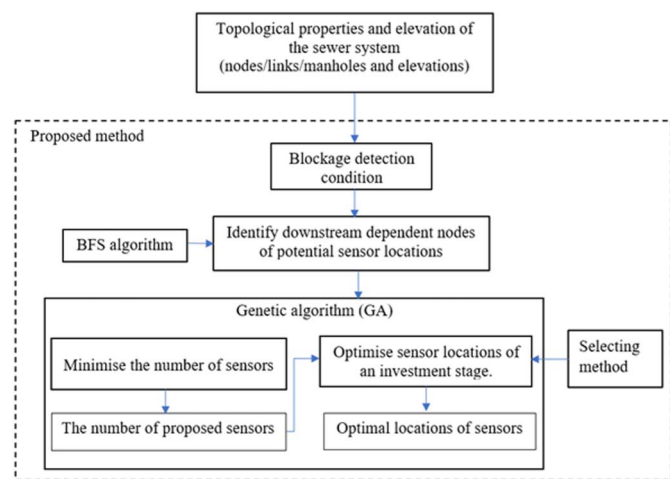


Fig. 5. Flow chart of the application of the proposed method.

The DEAP package of GA in Python is used to solve the optimization problems.

Results and Discussion

Identification of Downstream-Dependent Nodes

The results of identifying DD nodes for both sewer systems are presented in Fig. 6. A high coverage rate of manholes in Sewer System I can be seen [Fig. 6(a)]. Almost all manholes have at least one downstream node, except for Manhole 92 and the outlet of the network [Fig. 7(a)]. On average, a manhole in this system can have two or three DD nodes. The maximum number of DD nodes of a manhole is eight nodes (i.e., this manhole can cover seven links).

The number of downstream-dependent nodes for each manhole in Sewer System II [Fig. 6(b)] is very different from Sewer System I because of its steeper slope. In Sewer System II, only 47 out of 189 manholes have downstream-dependent nodes; i.e., the remaining 142 manholes do not have any downstream node. As a result, these locations should be required to place sensors to cover the entire system. In addition, the coverage rate of a manhole in the 47 manholes is low. Specifically, 36 manholes only have one DD node, and three manholes have a maximum of three DD nodes as shown in Fig. 6(b).

The variance as described previously partly reflects the impact of terrain conditions on the coverage rate of manholes. Due to the relatively flat terrain in Sewer System I, the elevation differences

between a manhole and its downstream nodes are relatively small. As a result, it is easy to identify DD nodes of the manholes in this system. This also leads to the overlap of DD nodes covered by different manholes in this system. In contrast, the sloping terrain has limited the coverage capacity of most manholes in Sewer System II.

Estimating the Minimum Number of Sensors for the Maximal Sewer Network Coverage

The minimum number of sensors required to cover the entire system is 64 and 170 for Sewer Systems I and II, respectively (Fig. 7). The optimally deployed sensors can cover almost all nodes in the systems, except for the uncovered upstream nodes that are not considered based on the proposed blockage detection condition (i.e., sensors must observe the water level rise at downstream nodes and pipes). In Sewer System I, the 64 sensors suggested by the GA can observe 224 downstream nodes out of the total 267 nodes, leaving only 43 nodes uncovered upstream. Similarly, in Sewer System II, the 170 recommended sensors cover 226 out of 323 total nodes.

A sensor in Sewer System I can detect water level changes at multiple downstream nodes, whereas most sensors installed in Sewer System II can only detect changes in water levels when blockages occur at the downstream pipe closest to them (Fig. 7). The difference in the minimum number of required sensors between the two systems and the detection capacity of a water level sensor highlights the significant influence of the terrain condition as discussed previously. Clearly, a system with a flat slope will demand a smaller number of sensors than a system with a steep slope. In such cases for steep sewer systems, relying on the water level rise data obtained from sensors may not be an effective way to assess the detection capacity.

It is also found that the sensor distribution with a minimum sensor number in the sewer system is not necessarily unique. In some cases, multiple solutions may have the same fitness but with different sensor locations. In practice, there may be different options for selecting sensor locations that lead to the same coverage in a local region. An example can be seen in Fig. 8 where either Manhole 101 or 23 in a branch of the sewer system can be selected for sensor deployment. This branch has eight manhole locations, and the result of the GA recommended six sensors to cover this branch. However, two optional location lists can be used in this situation: Manholes 73–13–15–19–101–24 and Manholes 73–13–15–19–23–24. Both options can cover all nodes and pipes in this branch.

This finding highlights the existence of several optimal choices for sensor placement within sewer systems. The presence of multiple optimal sensor locations may impact the selection of the best sensor

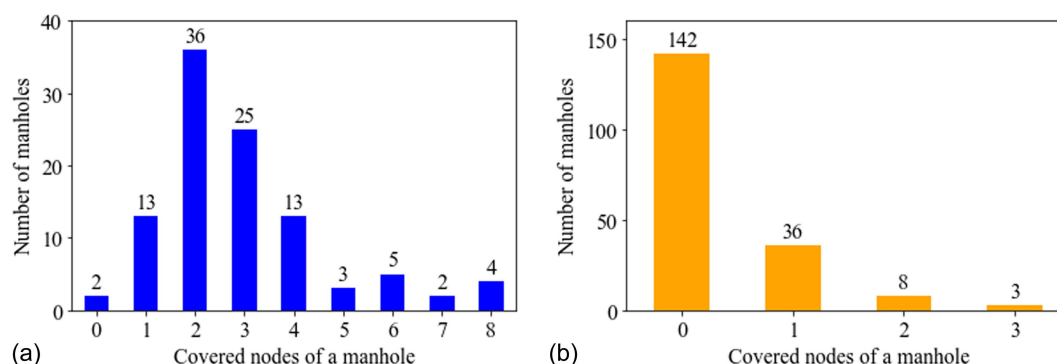


Fig. 6. Coverage capacity of manholes in sewer systems: (a) System I; and (b) System II.

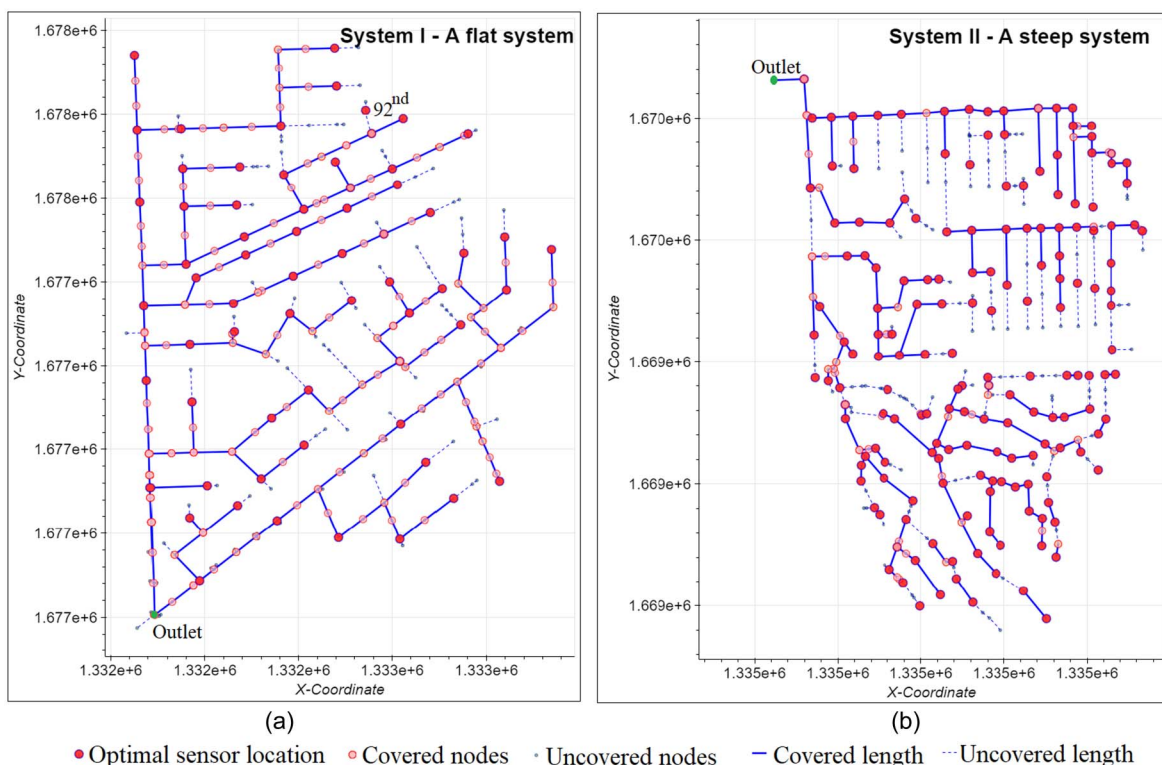


Fig. 7. Minimum sensor placements for (a) flat sewer system; and (b) steep system.

locations for various investment phases, as the optimal locations for subsequent phases could be influenced by those optimized in prior phases.

Optimal Locations for a Given Number of Sensors

An assumption is that only 10 available sensors are available for each installation in an ongoing investment phase for Sewer System I. The objective of the optimization problem is to identify optimal sensor locations to maximize the coverage of blockage detection using these 10 sensors. To evaluate the efficiency of the proposed method, the optimal sensor locations in Sewer System I are identified for two distinct scenarios: the first involves a new smart sewer

system with no sensors installed, and the second applies to an extension of the existing smart sewer system, where 10 sensors have already been installed.

Finding Optimal Sensor Locations for a New Smart Sewer System

In this scenario, the optimization is implemented to find the 10 best locations from 103 manhole locations in Sewer System I. The optimization outcome is validated by comparing the total pipe length covered by sensors in the 10 optimal locations with the total pipe length covered by the existing sensors. Fig. 9(a) displays the 10 optimal sensor locations obtained using the proposed method, while Fig. 9(b) represents the 10 existing sensor locations (black

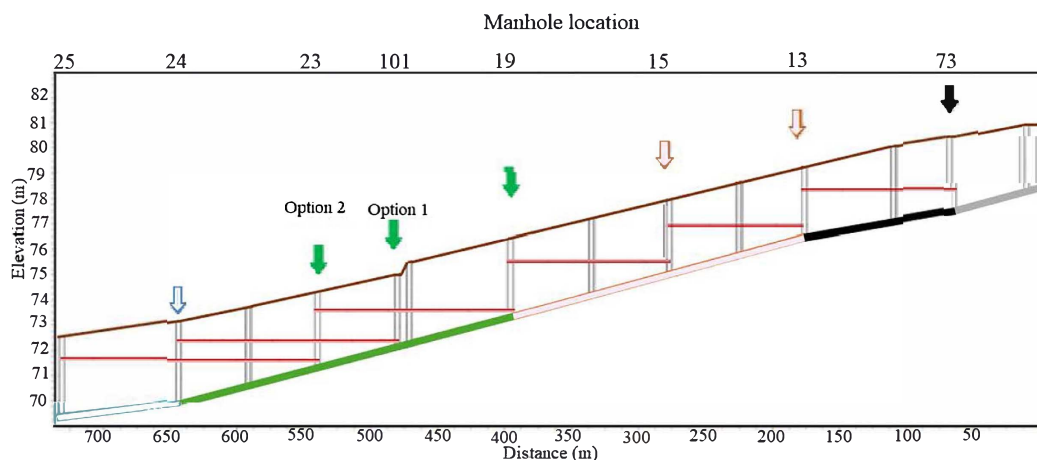


Fig. 8. Example of the optional manhole locations with the same length covered in a branch of Sewer System I. Option 1: Manhole 101 is selected; Option 2: Manhole 23 is selected.

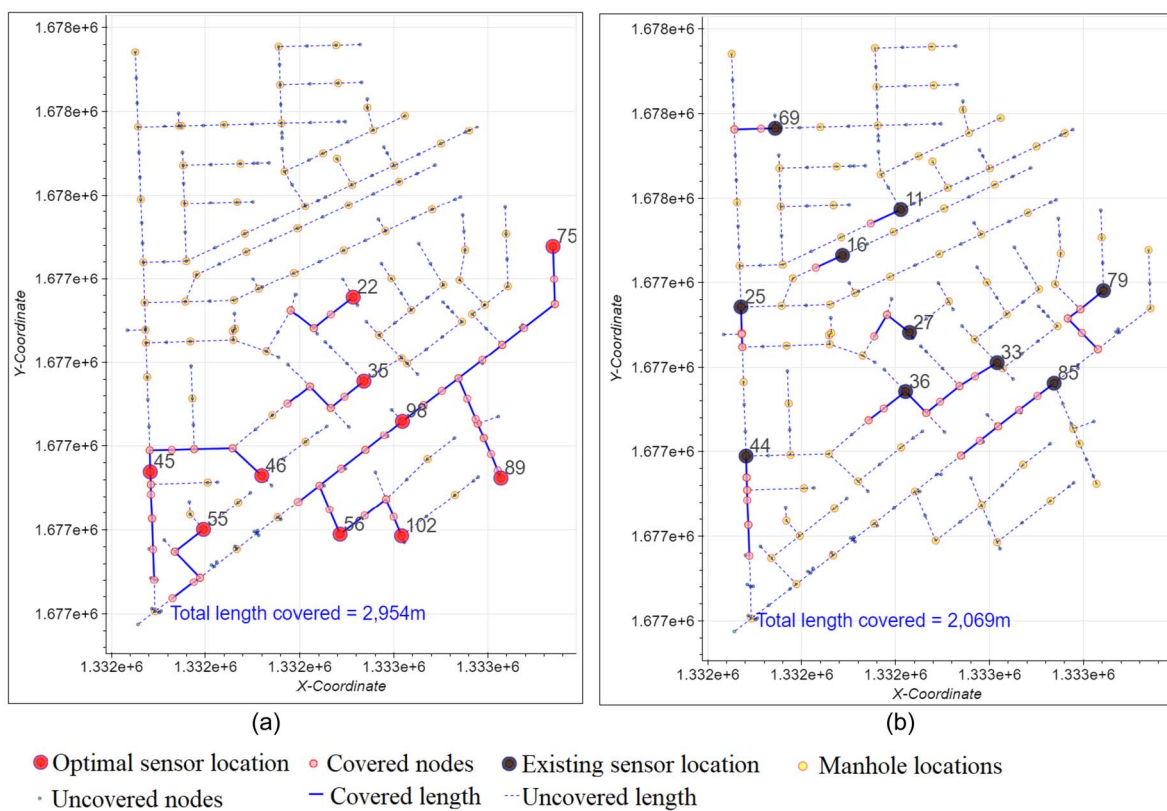


Fig. 9. Ten sensors installed in Sewer System I: (a) optimal locations; and (b) existing locations. Some optimal sensors can cover existing locations such as Sensor 22 covering Manhole 27.

circles). The existing sensors were strategically placed in maintenance holes based on historical data related to blockage events within the sewer system. It is noted that the optimal locations and existing locations are completely different. The coverage capacity of 10 sensors in optimal locations is better than the 10 existing sensors.

Considering the objective function, the total length covered by 10 sensors in the optimal locations is notably longer at 2,954 m compared to the coverage provided by the existing sensors, which totals 2,069 m. Although the allocation of the 10 existing sensor locations in Fig. 9(b) appears to have a well-spaced layout, it does not consider the topology of the network to maximize the effectiveness of each sensor. In contrast, the optimal sensor locations ensure that each sensor monitor at least three downstream pipes, whereas five of existing sensors (locations 11, 16, 25, 27, and 69) only observe one or two short downstream pipes as shown by the blue line in Fig. 9(b).

Moreover, **some optimal sensor locations enable the sensors to cover existing sensor locations, especially when the optimal positions are close to the existing ones.** For instance, the sensor in the optimal location 22 can observe the existing location 27 or optimal location 75 covers existing location 85. This highlights the suboptimal coverage of the existing sensors. In essence, the selection criteria prioritize the coverage capacity of sensors; the optimal locations provided by the proposed method are preferable for placing the initial 10 sensors over the existing locations. This implies that the proposed method effectively demonstrates its strong performance in identifying optimal sensor locations, even for the initial investment phase requiring the placement of a limited number of sensors. **This method can improve the effectiveness of sensor location distribution if considering other factors such as high-risk**

regions, environmental issues, or limiting the set of manholes to the main trunk lines to identify a wider coverage.

Finding Optimal Sensor Locations for an Extension of the Existing Smart Sewer System

It is important that relocating existing sensors can be challenging due to the high labor cost. If another group of 10 sensors is planned to be deployed in the next investment phase with the 10 existing sensors kept, the optimization problem now becomes looking for optimal locations for additional new sensors.

To solve this issue, the 10 existing sensor locations must be fixed before generating random individuals in the GA optimization process. This ensures that only the 10 new sensor locations are considered from the possible installation sites, while fitness evaluation still accounts for the set of 20 sensors. This approach can contribute to the reduction of the search space for the GA since the number of combinations when choosing 10 from the remaining 110 possible locations is significantly smaller than choosing 20 from 120 possible locations. Ultimately, the best individual corresponds to the minimum value of the fitness [satisfied Eq. (3)].

The locations of the 10 new sensors in Sewer System I can be installed in the Manholes 32, 46, 53, 55, 56, 61, 72, 75, 89, and 102 [Fig. 10(a)]. Considering the combination of these new sensors and the existing 10, the total pipe length covered by the collective set of 20 sensors is 4,555 m. While incorporating 10 new sensor locations with the existing 10 may align with the requirements of the investment phase, this combined set of 20 locations might not yield the most efficient solution for maximizing coverage within the sewer system. This limitation stems from the fact that the existing 10 sensor locations are not optimal. As illustrated in Fig. 10(b), optimizing both existing and additional sensor placements could extend total

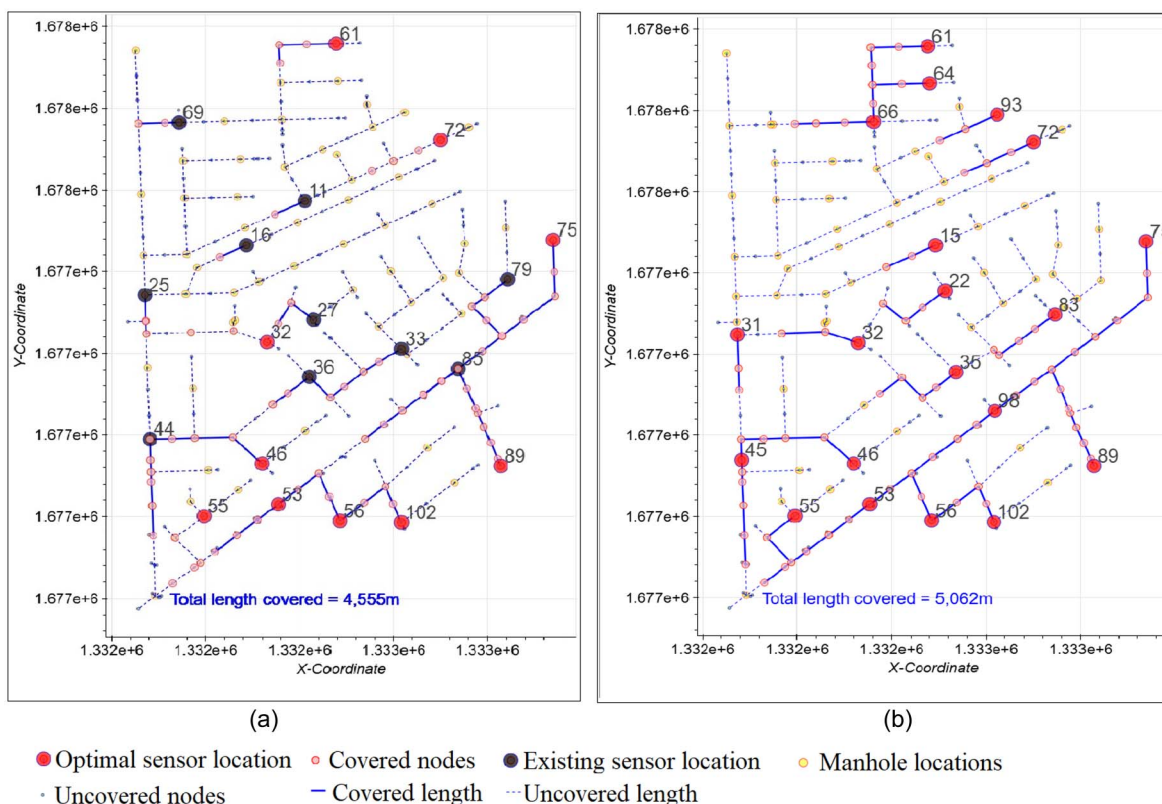


Fig. 10. (a) Ten new optimal sensor locations for the sensor network extension; and (b) all 20 sensor locations were optimized using the proposed method.

pipe coverage by 506.6 m, reaching 5,062 m. This enhancement results from selecting sensor locations concurrently under a global optimization approach. Thus, if the cost of relocating the 10 existing sensors is feasible, it would be an option to relocate these existing sensors and place all 20 sensors in the optimal locations, as suggested in Fig. 10(b).

Conclusion

A novel methodology, integrating a genetic algorithm with a blockage detection condition utilizing network topology and elevation data, has been used to determine the optimal number and placement of sensors for the smart sanitary sewer system. Within this methodology, a condition for identifying blockages at water level sensor locations is introduced, mandating that the sensor detects a rise in water level attributed to a blockage before overflow occurs at downstream nodes and pipes from the maintenance hole where the sensor is installed. The proposed method works well in both flat and steep sewer systems. However, its effectiveness is more pronounced in flat or moderately sloped systems, where elevation significantly impacts the observation capacity of water level sensors in adjacent manholes. In steep systems, the method can also yield good results if applied to less steep sections of the network or if the water depth threshold (H_d) is increased.

The application of the proposed methodology in case studies not only validates its ability to determine optimal sensor locations but also provides insight for the placement of a limited number of sensors required in a specific investment phase, either the initial phase or subsequent ones. The identified sensor locations can effectively cover the maximum length of pipes within the system.

This is attributed to the simultaneous selection method, which enables the identification of the optimal locations from a global optimization perspective, ultimately yielding the global optimum for the solution. In short, the proposed method is able to optimize both the number and placement of sensors across various investment phases to ensure complete coverage of the entire system, offering flexibility for water utilities in their investment decisions and thereby facilitating the realization of future smart sewer systems.

A limitation of this method is its current reliance only on the total length of covered pipes as the objective function for the optimization process. This method could be enhanced by integrating other relevant factors influencing optimal location selection as constraints within the optimization framework. In practice, factors such as historical data on blockage events, environmental factors, accessibility, and safety can play a role in sensor location selection. This broader consideration promises a more comprehensive and effective optimization process, and it will be considered in further studies. Moreover, relying solely on water level data makes it challenging to optimize sensor coverage in steep sewer systems. Other sensing methods need to be further investigated to enhance the monitoring coverage in steep sewer systems.

Data Availability Statement

The data of two smart sanitary sewer systems in this study are proprietary or confidential in nature and may only be provided with restrictions. Other data or code that support the findings of this study are available from the corresponding author upon reasonable request.

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